

2(a)(i)	1 <i>initial receptors</i> – osmoreceptors / hypothalamus ; 2 <i>coordination systems</i> – endocrine / hormonal ; 3 <i>final effectors</i> – <u>collecting</u>, duct / tubule ; 4 <i>response</i> – less <u>urine</u> produced / <u>urine</u> more concentrated / less watery <u>urine</u> OR more <u>water</u>, reabsorbed / taken up (into blood) OR (more) aquaporins added to <u>membrane</u> / <u>membrane</u> more permeable to water ;	4	1 IGNORE osmoregulators 2 IGNORE ADH 3 IGNORE posterior pituitary gland 3 CON other parts of nephron
2(a)(ii)	any two from: 1 if water potential, rises / high, cells, absorb water / swell / burst / lyse ; 2 if water potential, falls / low, cells, lose water / shrink / crenate / shrivel ; 3 water is a, solvent / medium / site, for, (named) reactions in cells / metabolic reactions ; 4 AVP ;	2	1 IGNORE flaccid / turgid / plasmolysed 1 ALLOW if too much water in blood / blood dilute, for 'high water potential' 2 ALLOW if too little water in blood / blood concentrated, for 'low water potential' ALLOW 1 mark for 'so cells do not burst or shrink' OR 'so cells do not gain water or lose water' if mps 1 and 2 not given 3 e.g. water needed for hydrolysis (reactions) in cells 4 e.g. control / maintain / important for, blood volume / blood pressure / hydrostatic pressure / glomerular filtration <i>ref.</i> prevent oedema
2(b)(i)	180 (g) ;	1	calculation = $200 \times 5 \times 0.18$

Question	Answer	Marks	Guidance
2(b)(ii)	any four from: <i>in, proximal convoluted tubule / PCT:</i> 1 (glucose) reabsorbed / moves, into, blood / (peritubular) capillary ; 2 Na⁺ leaves (epithelial / tubule) cell by active transport (at basal / basolateral, membrane) ; 3 glucose moves into cell by co-transport with Na⁺ (at luminal membrane) ; 4 glucose leaves cell by facilitated diffusion (at, basal / basolateral, membrane) ; 5 AVP ;	4	IGNORE ultrafiltration DO NOT ALLOW basement membrane for mps 2, 3, 4 but ECF after first error If Na⁺ / sodium ion, not stated CON once then ECF max 3 if PCT not stated CON if wrong part of nephron stated then ECF to max 3 ECF if cell not stated or implied (e.g. cytoplasm) for mps 2, 3 and 4 to max 3 2 ALLOW using ATP / pumped / by pumps, for 'active transport' 2 IGNORE transport protein 2 DO NOT ALLOW channel protein 3 ALLOW through co-transporter protein for 'by co-transport' 4 ECF from mp 2 if basement membrane repeated 4 ALLOW diffuses through GLUT, proteins / channels, for 'facilitated diffusion' 5 mp 2 establishes, Na⁺, diffusion gradient / concentration gradient e.g. secondary active transport